

Lab: Communicating Your Invention Across Audiences

Objective

Practice adapting your invention communication for different audiences by modeling what each audience already knows and what they need to learn. You will craft a pitch, a technical description, and a user instruction sequence, each tailored to align a different audience's generative model with your invention's core value.

Materials and Prerequisites

- Your invention idea from previous labs
- A listener (optional but highly recommended — a friend, family member, or classmate)
- Paper for sketching (or a digital drawing tool)
- A timer

Part 1: Modeling Your Audience (10 minutes)

Before you communicate, you must model the listener. What does each audience already know? What do they care about? What language do they use?

Audience 1: A potential investor or supporter (someone who has resources but no technical knowledge of your domain)

Aspect	Your Best Guess
What they already know about your problem domain	

Write your response here

| | What they care about most |

Write your response here

| | Technical vocabulary they are comfortable with |

Write your response here

| | What would make them skeptical |

Write your response here

| | What would make them excited |

Write your response here

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Audience 2: A technical collaborator (someone with expertise in a relevant field)

Aspect	Your Best Guess
What they already know about your problem domain	

Write your response here

| | What they care about most |

Write your response here

| | Technical vocabulary they are comfortable with |

Write your response here

| | What would make them skeptical |

Write your response here

| | What would make them excited |

Write your response here

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Audience 3: An end user (someone who will use your invention)

Aspect	Your Best Guess
What they already know about your problem domain	

Write your response here

| | What they care about most |

Write your response here

| | Technical vocabulary they are comfortable with |

Write your response here

| | What would make them skeptical |

Write your response here

| | What would make them excited |

Write your response here

Part 2: Crafting a 60-Second Pitch (15 minutes)

Write a pitch for Audience 1 (the investor/supporter). Your pitch should follow the model alignment structure:

Step 1 — Create the prediction error (the problem): Write one sentence that reveals a surprising problem.

Write your response here

Step 2 — Present the solution: Write 2-3 sentences explaining what your invention does and why it works.

Write your response here

Step 3 — Establish value: Write one sentence connecting your invention to what this audience cares about.

Write your response here

Step 4 — Establish credibility: Write one sentence explaining why you are the right person to build this.

Write your response here

Full pitch (combine steps 1-4 into a flowing paragraph):

Write your response here

If you have a listener, deliver the pitch out loud. Then answer:

What questions did the listener ask?

Write your response here

What parts of your pitch did they seem to understand immediately?

Write your response here

What parts required additional explanation?

Write your response here

What would you change for next time?

Write your response here

Part 3: Writing a Technical Description (10 minutes)

Write a technical description for Audience 2 (the collaborator). This should externalize the causal model of your invention.

Mechanism description (how the invention works, using appropriate technical vocabulary):

Write your response here

Key parameters (specific values, ranges, or specifications):

Parameter	Value/Range	Why it matters

Write your response here

Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

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Open questions (what you do not yet know and would want a collaborator to help with):

Write your response here

How does this description differ from the pitch you wrote in Part 2?

Write your response here

Part 4: Designing User Instructions (15 minutes)

Design a step-by-step instruction sequence for Audience 3 (the end user). Follow the IKEA principle: build the user's model incrementally, one step at a time.

Step 0: What the user sees before they start (the initial state):

Write your response here

Instructions (write 5-8 steps, each with one action and one observable result):

Step	Action (what the user does)	Result (what they observe)	Why this step matters
1			

Write your response here

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Write your response here

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Write your response here

|| 2 |

Write your response here

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Write your response here

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Write your response here

|| 3 |

Write your response here

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|| 4 |

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Write your response here

|| 8 |

Write your response here

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Write your response here

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Write your response here

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Common mistakes (what might go wrong if the user's model is not correctly calibrated):

Write your response here

Affordances (features of the design itself that communicate how to use it without written instructions):

Write your response here

Part 5: Reflection on Communication as Active Inference (10 minutes)

How did your message change across the three audiences? What stayed the same, and what transformed?

Write your response here

Which audience was hardest to communicate with? What made it hard — the model gap, the vocabulary gap, or the priority gap?

Write your response here

What is the biggest gap between your understanding of your invention and what you can currently communicate? What knowledge is still trapped in your head and has not been externalized?

Write your response here

If your invention communicated its own function through its design (affordances), which written instructions could be eliminated?

Write your response here

Discussion and Debrief

1. **Model inference:** How accurately did you model each audience? If you delivered the pitch to a real person, did their questions reveal that your model of their knowledge was wrong in specific ways?
2. **Compression vs. completeness:** The pitch compressed your model heavily; the technical description expanded it. At what point does compression lose essential information? At what point does completeness overwhelm the listener?
3. **The demonstration question:** Which aspects of your invention would communicate best through demonstration rather than description? If you had the chance to show rather than tell, what would you show?
4. **Affordance design:** How many of your written instructions could be eliminated if the invention's design itself communicated its function? What design changes would create clearer affordances?

5. Communication as invention: In what sense is the communication artifact (the pitch, the document, the instruction) itself an invention — a designed system with a purpose, constraints, and iterative improvement potential?

Write your response here